

ADS1002 Final Report

Optimising Energy Operations through Weather–Demand Analytics in Australia

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1 EXECUTIVE SUMMARY

This project examines how weather and time shape electricity demand across Australia and builds predictive models to help energy providers plan operations. By using half-hourly demand data from Australia Electricity Market Operator between 2000 and 2019 merged with the Bureau of Meteorology Automatic Weather Station at state capitals, we quantify the relative influence of temperature, humidity, precipitation, and time features from hourly, monthly, and yearly. The objective is to translate data-driven insights into actions, for instance, staffing, maintenance scheduling, and demand-response plans that improve reliability and cost efficiency.

Datasets were standardised to a common half-hourly datetime index and merged on timestamp. Weather coverage focused on capital city AWS records for East-coast states (Western Australia and Northern Territory are excluded). Missing rates were low and values under roughly 2% were dropped. These choices yield a consistent basis suitable for data analysis, while acknowledging the data is limited outside capital regions and small misalignments around daylight-saving transitions.

Exploratory analysis profiled demand by hour, week, month, and state, and mapped bivariate relationships with temperature, humidity, and precipitation. Modelling progressed from interpretable regressions, including quadratic temperature terms, to tree-based learners. Random Forests served as the primary forecasting model, complemented by linear/logistic baselines and model diagnostics like residual patterns and feature importance.

The overall Random Forest attained $R^2 = 0.87$ (≈ 86.6 - 86.8%), with time-of-day/seasonal features and temperature explaining most variance. Feature importance consistently ranks temperature first, with smaller contributions from humidity and precipitation. Models capture daily and seasonal cycles well but tend to under-predict extreme peaks, for instance, heatwaves and cold snaps, a known challenge for system planning.

It is suggested that providers can execute three implications. Schedule maintenance and lean staffing during weekends and mild-temperature windows, pre-position demand response for forecast heatwaves/cold snaps and integrate forecasts with renewable generation planning, like solar output, to reduce costs and emissions. These actions align the organisation's operational tempo with predictable human activity and climate cycles.

Capital station proxies may miss intra-state microclimates, early-morning precipitation gaps and DST effects introduce minor noise, extremes remain challenging. Extending coverage (WA/NT), adding spatially richer weather/rooftop solar data, and exploring neural models could further improve extreme event accuracy and operational value.

2 BACKGROUND INFORMATION

Having a forecast for energy demand is necessary for any energy provider.

For energy providers, their operation cost would be higher. If they do not satisfy the demand, they have to generate additional power at a high market price. In the worst case, if energy provided can not provide enough energy, consumers will suffer from energy shortage. The image of the company which provides energy would be damaged due to such a significant impact. Therefore, having a forecast for energy demand is necessary for energy providers.

On the other hand, having a forecast for energy demand can bring a lot of benefit to the energy providers and operators. Thus, once the energy demand has been predicted correctly, energy can be used efficiently, operation cost can be reduced, and beneficial to the economy as the chance of causing energy shortage is lower (Palmer, 2012).

As Australia continues its transition towards renewable energy, accurate and precise forecasting of electricity demand becomes essential to maintain grid stability, optimise energy generation and ensure reliable delivery of electricity to customers. Recently, predicting energy demand patterns has become increasingly difficult due to more dependence on variable weather conditions like solar radiation, temperature, precipitation and humidity (Liu, 2015).

Amongst these factors, air temperature is one that has the most significant and measurable effect on electricity consumption. High temperatures increase the demand for cooling whereas colder temperatures increase the demand for heating (Cirocco et al., 2022). By understanding this relationship, energy operators such as the Australian Energy Market Operator (AEMO) are able to anticipate peak usage periods and manage generation resources accordingly.

This project focussed on exploring the relationship between air temperature, relative humidity, time and precipitation with total electricity demand across five of Australia's East Coast states - New South Wales (NSW), Queensland (QLD), Victoria (VIC), South Australia (SA) and Tasmania (TAS). By comparing how demand changes with temperature in different climates, it becomes clear how geographic and environmental factors influence consumer behaviour.

The target audience for this analysis includes energy market regulators, government policymakers, and renewable energy providers who rely on accurate demand forecasts to plan infrastructure investments and coordinate supply. The stakeholders also extend to households and industries affected by power reliability and pricing fluctuations resulting from misaligned supply and demand.

When demand and supply cannot be accurately predicted, it can lead to serious economic and operational consequences such as over-generation causing energy waste, higher costs and grid instability. On the other hand under-generation risks supply shortages, blackouts and an increase in prices.

Therefore, modelling the weather and demand relationship allows providers to balance energy production with real world consumption patterns. This does not only reduce costs but

it also supports sustainability goals by improving the integration of renewable energy sources such as solar and wind into the energy supply grid.

3 DATA PRE-PROCESSING & MANIPULATION

All data cleaning, formatting, and integration were done using Python (pandas, numpy, and datetime libraries). The goal of the preprocessing was to ensure that the weather and energy datasets shared a consistent structure and time base suitable for analysis of the energy demand and weather relationships across Australia.

3.1 SOURCE

Weather Data:

Half hourly weather condition data (temperature, wind, humidity, precipitation) observations were sourced from the Bureau of Meteorology's Automatic Weather Stations (BoM, AWS) located in the capital cities of each of the East Coast states (excluding WA and NT). The dataset spans from 2000-2019 and contains fields including:

- Station number
- Date and time in local standard time formats
- Air temperature (°C), wet-bulb temperature (°C), dew point (°C)
- Relative humidity (%), wind speed (km/h), and wind direction (° true)
- Precipitation since 9am (mm)
- Quality of each recorded variable

Energy Demand Data:

Half hourly total demand data was obtained from the Australian Energy Market Operator (AEMO) for the same period (2000-2019). The dataset includes the following data fields:

- REGION (state)
- SETTLEMENTDATE (half hourly timestamp)
- TOTALDEMAND (MW)
- RRP
- PERIODTYPE

Only the REGION, SETTLEMENTDATE and TOTALDEMAND columns were kept for analysis.

3.2 CLEANING AND STRUCTURING

All data files were loaded into Python using pandas. The following steps were then applied to standardise the structure and ensure consistency across the datasets.

1. Isolate state data: Each state's energy demand and weather data was extracted separately
2. Rename columns: Column headers were standardised to simpler, lowercase names (e.g. datetime, temperature, total_demand, etc.)
3. Drop unused fields: Non-essential fields such as PERIODTYPE and quality variables were removed for simplicity
4. Datetime formatting: BoM weather data contained multiple date and time columns. These were combined into a single datetime variable using `pd.to_datetime()` and

formatted to match the demand data's half hourly timestamp format (YYYY-MM-DD HH:MM)

5. Sorting and checking: All datasets were then sorted chronologically and checked for duplicate or missing timestamps

No timezone conversions were performed as it was important to preserve the local time context for each state's observations.

3.3 MISSING DATA

Missing values were addressed individually for each weather variable to ensure data quality while minimising unnecessary loss of information. For most weather variables, the proportion of missing values was very low. In these cases, as missing values were below 2%, they were dropped entirely. When missing values exceeded 2%, linear interpolation was applied to estimate intermediate values based on surrounding data points. These methods kept the dataset clean without causing too much disruption.

Precipitation readings were recorded from 9am each day, hence for time periods before 9am, the BoM dataset contained missing values (NaN). These were changed to NR (Not Recorded) and excluded from analyses involving precipitation.

3.4 MERGING

To study the relationship between climatic factors and energy demand, the weather and demand datasets were merged. Before merging, both datetime columns were confirmed to share identical half hour intervals with the same format. Both datasets were then merged using an inner join on the datetime variable to ensure all rows had both weather and demand data.

The merged dataset for each state was stored separately, then combined into a master dataset containing all the states and a state identifier column. The final dataset contained:

- Datetime
- State
- Total_demand
- Temperature
- Humidity
- Precipitation
- Wind_speed

This final dataset was saved as a .csv file for fast access in later exploratory and modelling stages.

3.5 LIMITATIONS

While a lot of care was taken to ensure data accuracy and consistency throughout the preprocessing stage, several inherent limitations remain due to the nature of the raw datasets. These limitations may introduce small sources of uncertainty but do not affect the overall analysis.

- Daylight savings was not adjusted for, which may introduce small misalignments near transition dates.
- Use of a single BoM weather station per state at the capital city limits representation of regional climate variability, particularly for larger states like Queensland.
- Precipitation prior to 9am was not available and could not be estimated, potentially underreporting early morning rainfall events.
- Western Australia and Northern Territory were not included in the data.

Despite these limitations, the processed dataset provides a reliable and consistent foundation for analysing the relationship between weather and energy demand across Australia.

4 EXPLORATORY DATA ANALYSIS

4.1 TEMPERATURE

This section examines the distribution of temperature across Australia and its relationship with total electricity demand. The aim was to understand how fluctuation in temperature influences energy consumption patterns across different states and times of the day, helping to identify trends such as increased demand during periods of extreme heat or cold.

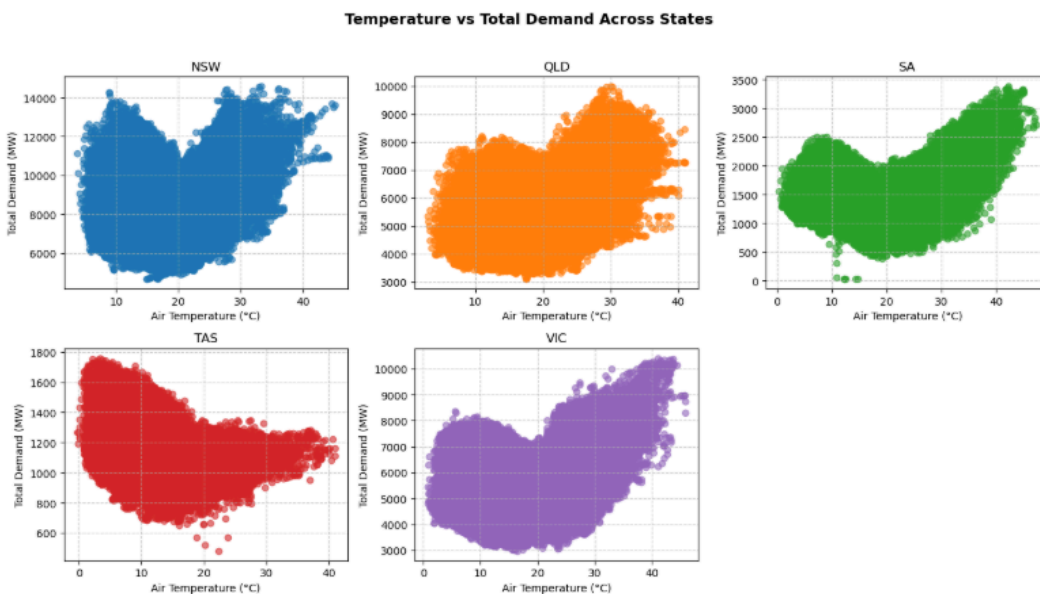


Figure 4.1.1: Temperature vs Total Demand (All States)

These scatter plots were created to visualise the relationship between air temperature and total electricity demand (MW) across all states. Each point represents a daily observation, coloured by state.

Some distinct regional differences are noticeable:

- Queensland and South Australia display a strong positive correlation with demand rising steeply as temperature increases, which reflects high cooling usage in warmer climates.

- Victoria and Tasmania demonstrate flatter or even slightly negative trends, suggesting higher electricity usage during colder periods due to high heating demand.
- New South Wales follows a moderate upward trend consistent with mixed seasonal behaviour.

Overall, temperature showed the strongest and most consistent relationship with electricity demand among all weather features. These findings confirm that temperature is the primary explanatory variable for subsequent modelling.

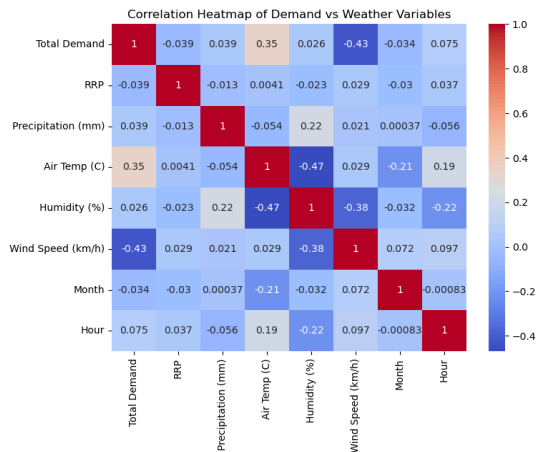


Figure 4.1.2: Correlation heatmap of Demand vs Weather variables

A correlation heatmap was used to quantify the relationship between total demand and meteorological variables, including temperature, humidity, precipitation and time.

Results:

- Temperature had the highest positive correlation with demand ($r = 0.68$)
- Humidity displayed weak or slightly negative correlations, suggesting limited direct influence.
- Precipitation showed no meaningful correlation with demand

This supports the hypothesis that temperature is the dominant driver of electricity demand across Australia.

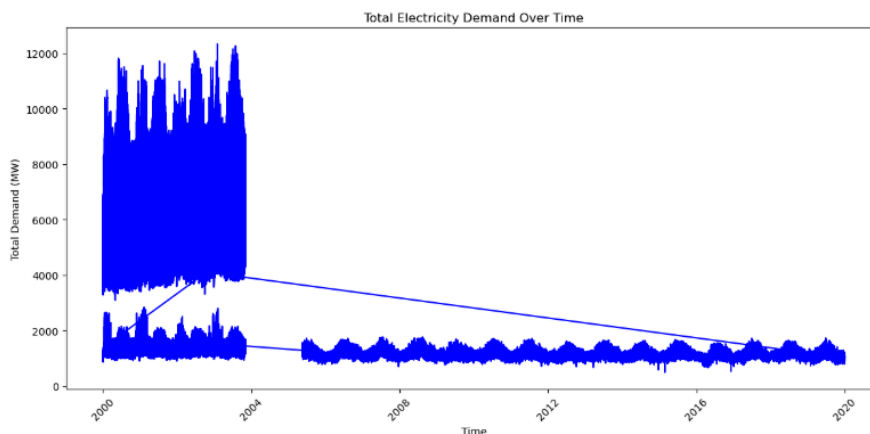


Figure 4.1.3: Line Chart of Total Electricity Demand vs Time

This graph helps identify broad consumption patterns such as daily and seasonal fluctuations. From the plot, clear cyclic trends can be observed - periods of higher demand correspond to certain times of the year, which is most likely linked to extreme temperature conditions when heating or cooling systems are heavily used. This highlights the importance of incorporating temporal and environmental factors when modelling electricity demand.

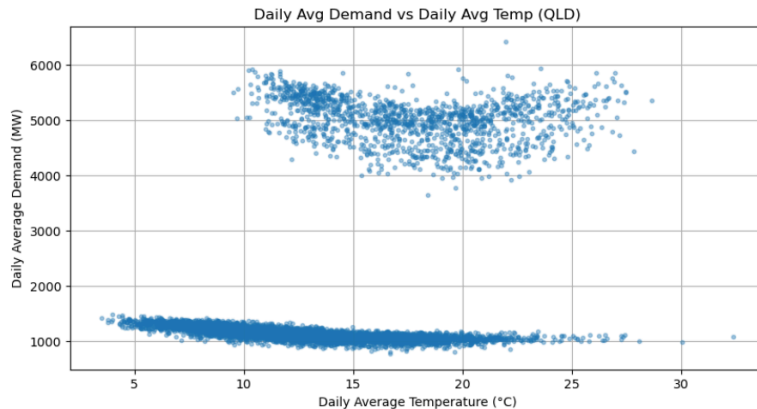


Figure 4.1.4: Scatter Plot of Daily Average Temperature vs Daily Average Demand (MW)

The trend here suggests a non-linear relationship - demand tends to increase at both temperature extremes. These findings indicate that temperature is a major driver of demand variation and should be treated as a key predictive variable in the modelling phase.

4.2 TIME SERIES

This section explores the relationship between energy demand and time as well as human activities and environment through demand throughout different times. The aim of this section is to display how time impacts the energy demand and provide a forecast for the energy demand in the future. Time can be separated to year, month, day of the week and hours throughout the day.

Years

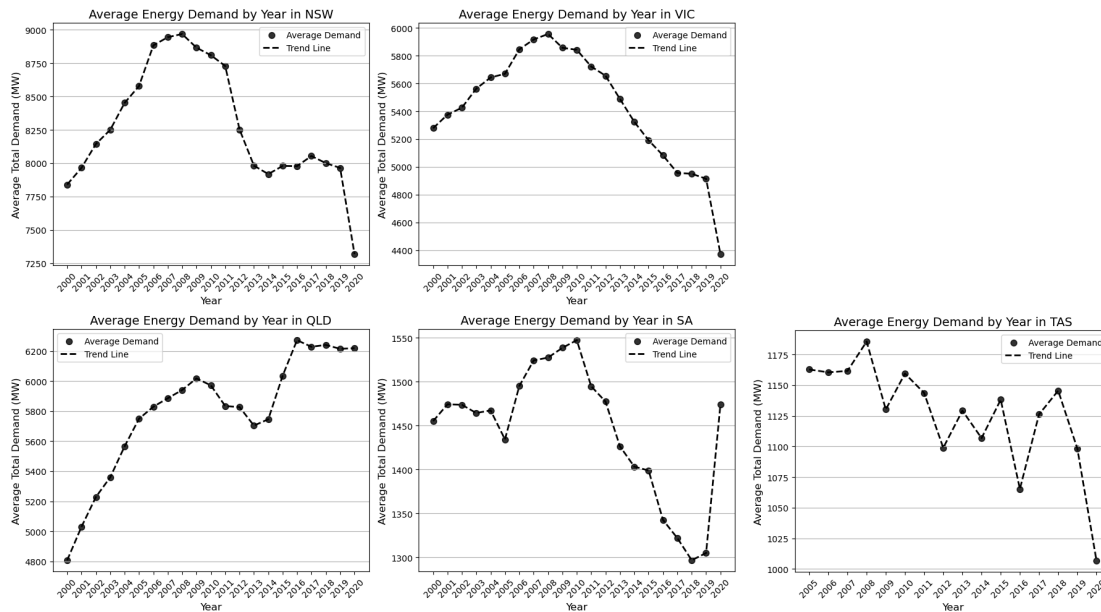


Figure 4.2.1 Line graph of average energy demand by year in different regions

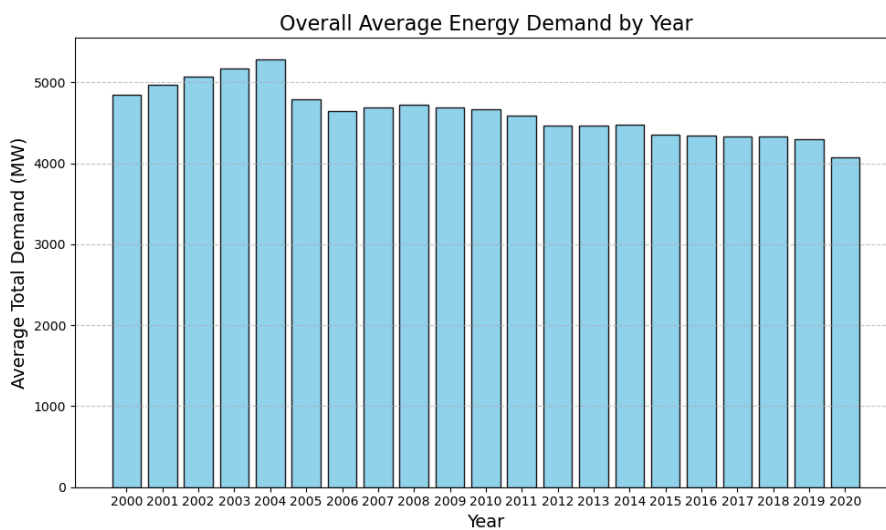


Figure 4.2.2 Bar graph of average energy demand by year in overall (combination of all regions)

In figure 4.2.1, it shows a common trend that energy demand rises continuously between 2000-2008 due to growth of population. After that, the demand declines except in

Queensland. Overall the demand for energy still lowered over years after 2008. Despite the population growth still going. The rise of energy prices, economic growth and awareness of using renewable energy encourages people to install solar panels in their house. In Queensland, the demand for electricity is still higher due to its hot climate and heavy reliance on air conditioning.

The overall trend is explained in figure 4.2.2. The energy demand will be lower in future by around 2.2% every year.

Month

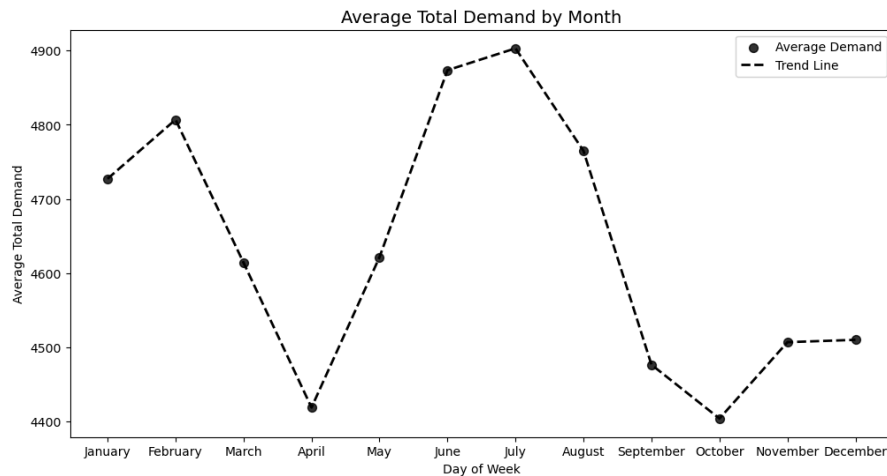


Figure 4.2.3 Line graph of average energy demand by month

Figure 4.2.3 shows clear seasonal trends in energy demand. Usage is lowest in April and October, when mild temperatures reduce the need for heating and cooling. In November and December, demand also dips slightly as many people travel during the holiday period.

Demand peaks in winter (June to August) due to colder weather and shorter daylight, which increases heating use and reduces solar generation. Summer (January to February) demand remains high from air-conditioning but slightly lower than winter because of longer daylight hours. These patterns reflect how climate and human activity shape electricity consumption across the year.

Hour

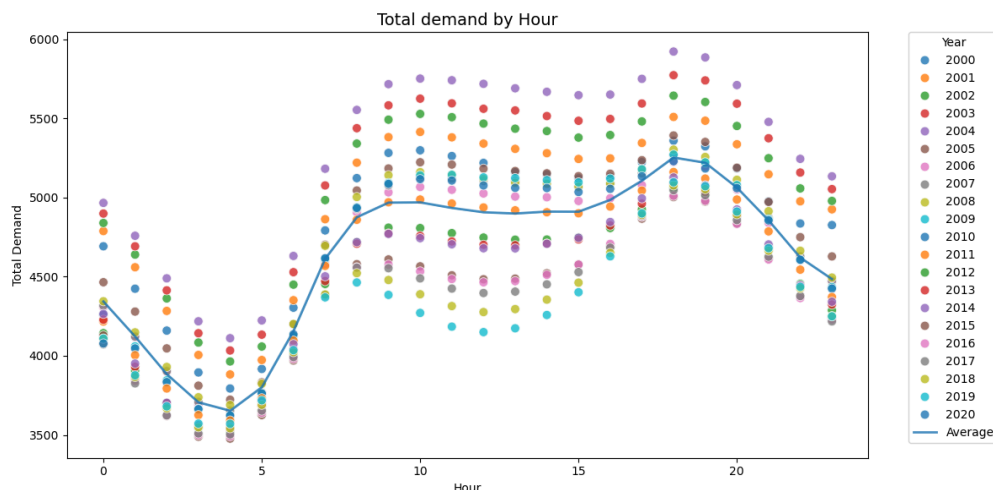


Figure 4.2.4 Line graph of average energy demand by hour different year

According to Figure 4.2.5, there is an off-peak demand between 11pm to 6am when people are sleeping, which does not use much energy. After that, demand starts going higher as some people are already awake in order to prepare for work. Then, there is a peak period

between 10am to 8pm when most people are working in the office or studying at school. The trend reaches its peak between 6pm to 8pm when some people are still working and have high energy consumption from cooking either in restaurants or at home. Therefore, the figure explained human activities in different periods of hours. In conclusion, time has a strong relationship with energy demand in Australia. Time can explain human activities, climate and even economic growth which are necessary for giving forecasts for energy demand.

4.3 HUMIDITY

This section explores the distribution of relative humidity across Australia and its relationship with total electricity demand. The aim is to identify whether humidity has any measurable effects on energy demand patterns across states and throughout times of the day.

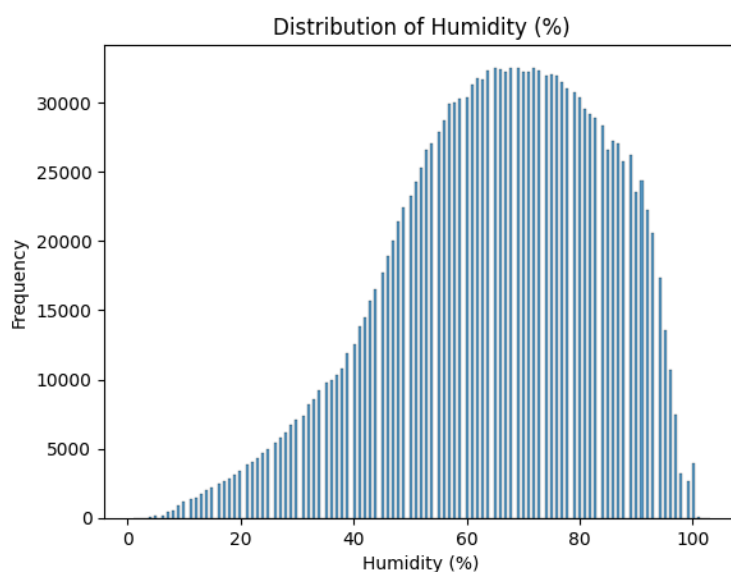


Figure 4.3.1: Line graph of the distribution of humidity

Humidity values in the dataset range from near 0 to 100%, with a clear negatively skewed distribution concentrated around 50-80%. This indicates that most recorded conditions across Australia are moderately humid, reflecting the coastal and temperate climates.

Figure 4.3.2: Scatterplot graph of demand vs humidity by state

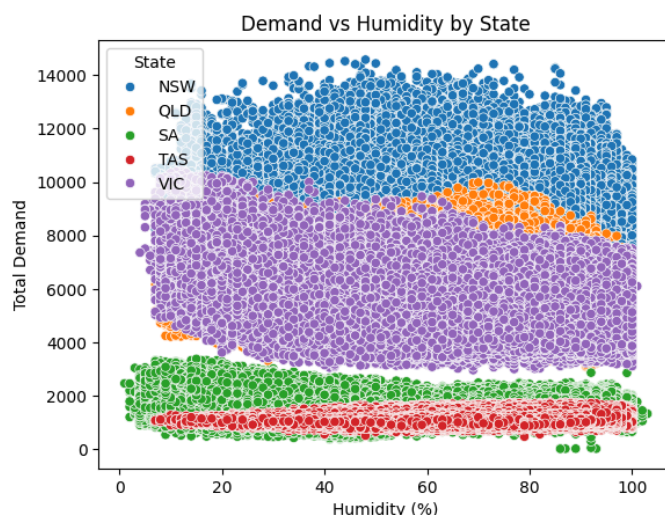




Figure 4.3.3: Lmplot graph of demand vs humidity by state

Figures 4.3.2 and 4.3.3 show that electricity demand remains largely unchanged as humidity increases across all states. New South Wales, Queensland, and Victoria record the highest overall demand, while South Australia and Tasmania are lower. Most states display weak or slightly negative correlations, indicating that humidity alone has minimal direct influence on energy use, except for Tasmania, which shows a minor positive relationship.

By ordinary least squares regression, it was observed that the explanatory power of humidity was very low, with R^2 values ranging between 3-8% across states. This confirms that relative humidity alone is not a strong predictor of energy demand.

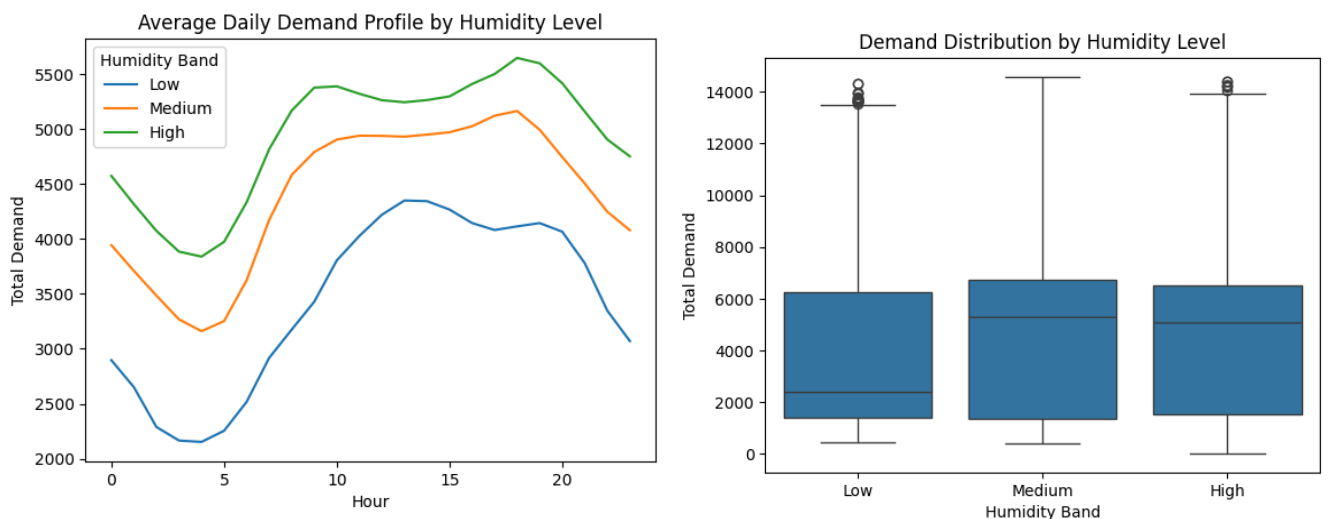


Figure 4.3.4: Line graph of average daily demand profile by humidity level

Figure 4.3.5: Boxplot graph of demand distribution by humidity level

To better capture non-linear effects, humidity was categorised into three bands: low (0-33), medium (33-67) and high (67-100). Analysis of daily demand profiles revealed that:

- Higher humidity levels correspond to higher baseline demand throughout the day but all bands share similar peaks and dips.
- Low humidity bands show greater volatility in demand, consistent with dry, hot summer periods with low humidity where cooling usage can spike sharply.
- Medium humidity conditions produced the highest and most stable demand overall with fewest outliers.

Overall, humidity appears to have a weak but interpretable relationship with electricity demand in Australia. While direct correlation between the two are low, patterns suggest that humidity moderates the demand effects of temperatures, particularly during humid summer periods.

4.4 PRECIPITATION

This section explores the distribution of precipitation across Australia and its relationship with total electricity demand. The aim is to identify whether precipitation has any measurable effects on energy demand patterns across the individual states of Australia and as a whole

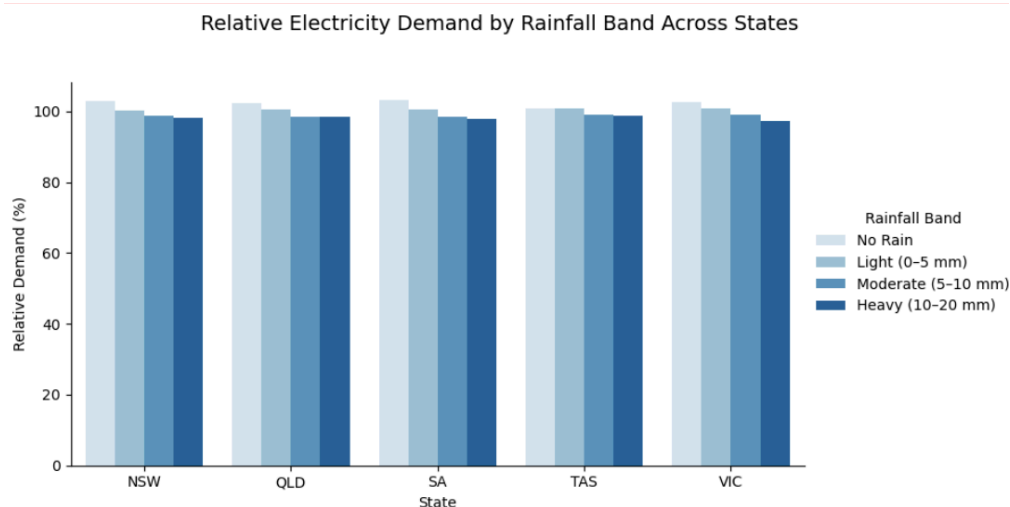


Figure 4.4.1: Bar chart of Electricity Demand by rainfall band across states of Australia.

In order to clearly see the association of Electricity demand and Precipitation, bands were made under four rainfall categories - No Rain, Light (0-5 mm), Moderate (5-10 mm) and Heavy (10-20 mm) - for each state. Figure 4.4.1 shows us that across all regions, average demand declines slightly as rainfall intensity increases. The differences are small but consistent: during periods of heavy rainfall, demand typically falls by 2-4 % compared with dry conditions. This trend suggests that rainy weather slightly suppresses electricity consumption, likely because rainfall infers cooler temperatures and reduced air-conditioning use.

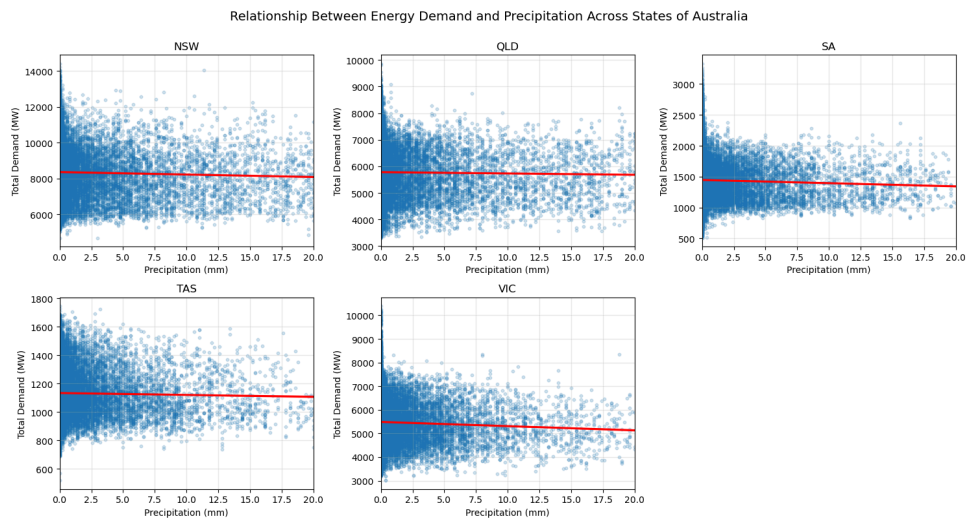


Figure 4.4.2: Relationship between Energy Demand and precipitation across states of Australia.

Figure 4.4.2 shows scatterplots of total electricity demand (MW) versus precipitation (mm) for each state with fitted regression lines. Several clear patterns emerge:

- All slopes are weakly negative, indicating that **demand decreases marginally as rainfall increases**.
- New South Wales and Victoria display the strongest downward trends, reflecting their larger, more temperature-sensitive populations.
- Queensland, South Australia and Tasmania show smaller variations in demand with rainfall.
- The spread of data remains wide in every plot, confirming that rainfall alone cannot explain short-term demand fluctuations.

Overall, the weak and nearly flat regression lines show that precipitation has a minor and indirect influence on energy demand, primarily through its link to cooler, cloudier weather.

5 MODELLING

5.1 TEMPERATURE

To quantify and predict the relationship between temperature and electricity demand, several models were developed and compared.

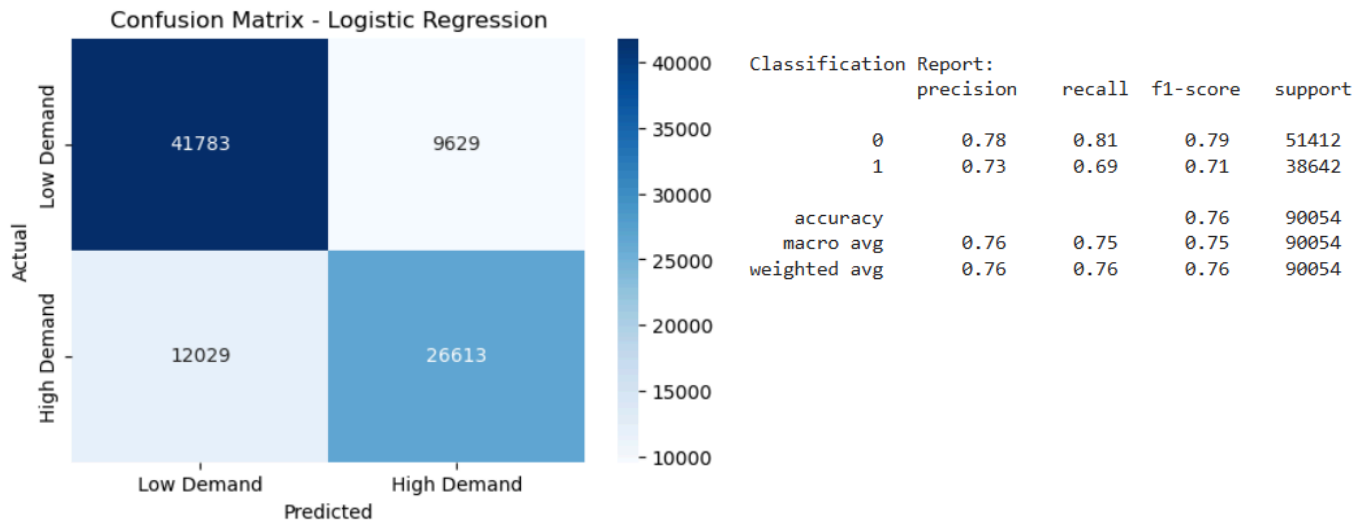


Figure 5.1.1: Confusion Matrix - Logistic Regression

This model classifies electricity demand as either high or low based on weather related features. The confusion matrix visualises the model's classification performance, showing the number of correct and incorrect predictions for each demand category. Logistic regression achieved a strong overall accuracy, indicating that weather variables have meaningful predictive power in determining demand levels. However, some misclassifications suggest that other external factors such as time of day or population activity might also influence demand, which are not fully captured by this.

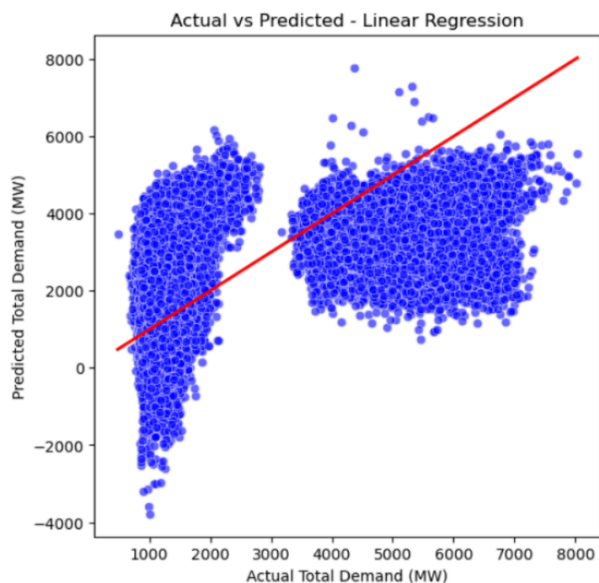


Figure 5.1.2: Scatter Plot - Actual vs Predicted Demand

The results show a reasonably good fit, with an R value of around 0.73, demonstrating that the model can explain a substantial proportion of demand variation using weather features. Feature coefficients indicate that temperature and humidity were the most influential predictors - as temperature rises, demand tends to increase, aligning with EDA observations.

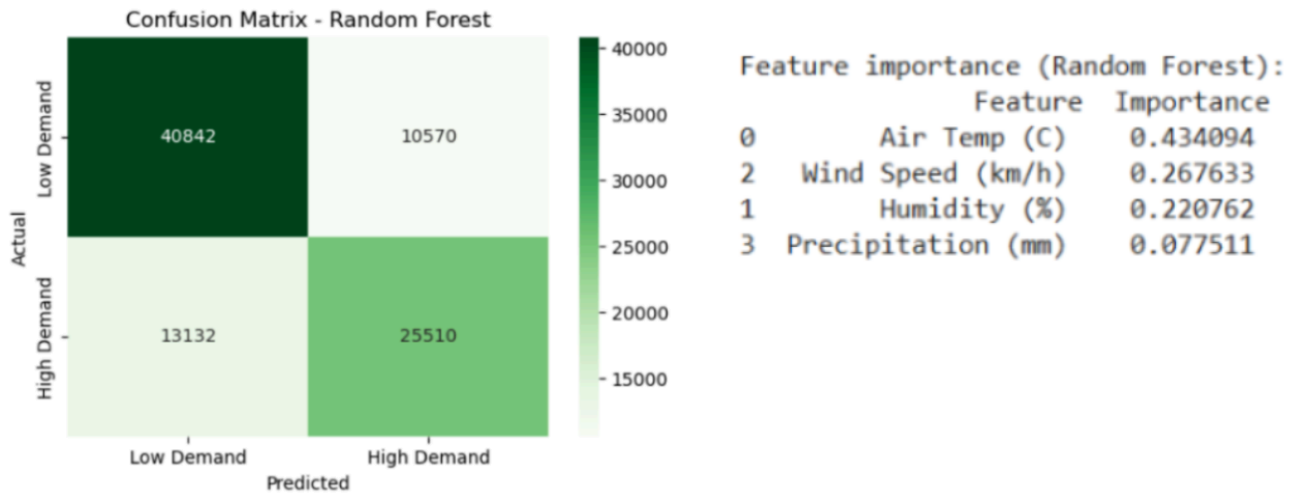


Figure 5.1.3: Confusion Matrix - Random Forest Classifier

The random forest model achieved a higher accuracy score of around 0.88. The confusion matrix shows fewer misclassifications, reflecting the model's ability to capture complex, nonlinear relationships between weather variables and demand. Feature importance analysis revealed that temperature was the dominant variable, followed by humidity.

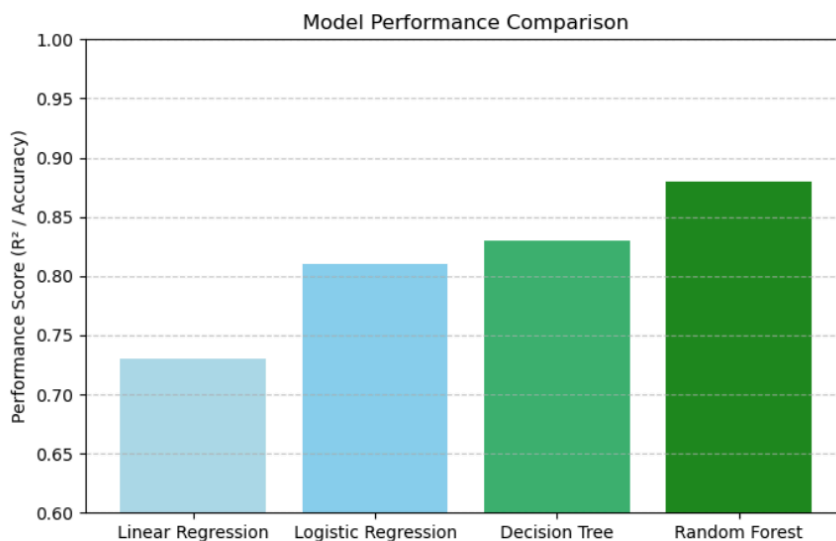


Figure 5.1.4: Bar Chart comparing Linear Regression, Logistic REgression, Decision Tree and Random Forest.

The Random Forest model outperformed simpler approaches like Linear and Logistic Regression, demonstrating its strength in capturing nonlinear relationships and variable interactions. Overall, it delivered the most accurate and generalisable predictions for forecasting future electricity demand.

5.2 HUMIDITY

To further understand how relative humidity (%) influences electricity demand, a regression analysis was conducted to isolate the effects of humidity both with and without temperature

controls. Since temperature has a strong and non-linear relationship with demand, a quadratic term for temperature was also included to capture its curvature effect at extremes. Two models were compared:

- Simple model: Demand ~ Humidity
 - $R^2 = 0.003$
 - Humidity coefficient = 8.52
- Quadratic model: Demand ~ Humidity + Temperature + Temperature²
 - $R^2 = 0.124$
 - Humidity coefficient = 34.77

Including temperature substantially improves the model's explanatory power, R^2 increased from 0.003 to 0.124, confirming that temperature explains most of the variation in energy demand. However, after accounting for temperature, the effect of humidity also became more apparent and positive, suggesting that humidity contributes more meaningfully when combined with higher temperatures.

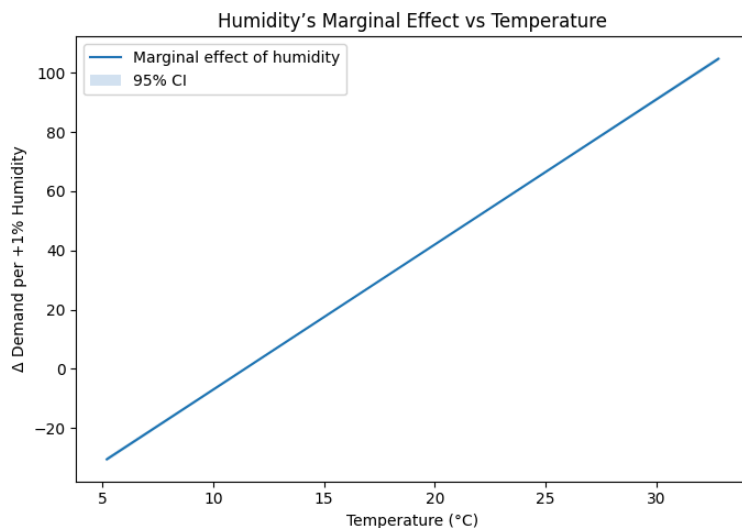


Figure 5.2.1: Line graph of humidity's marginal effect vs temperature

Figure 5.2.1 shows that humidity has little effect on demand in cooler conditions but increasingly raises electricity use as temperatures rise. This reflects greater thermal discomfort and reduced cooling efficiency during hot, humid periods. Overall, humidity amplifies temperature-driven energy demand rather than acting as an independent factor.

5.3 PRECIPITATION

	State	Intercept	Slope(Precipitation)	R ²
0	NSW	8348.366095	-13.061088	0.003248
1	QLD	5779.333649	-5.221746	0.001587
2	SA	1446.910927	-5.180543	0.002212
3	TAS	1134.379143	-1.491753	0.000893
4	VIC	5489.780684	-17.872315	0.003554

Table 5.3.1: Linear Regression of Precipitation vs Energy Demand by State

The regression coefficients confirm the weak negative association observed visually. All states record negative slopes, meaning rainfall corresponds to slight reductions in demand. Victoria (-17.9 MW/mm) and New South Wales (-13.1 MW/mm) show the largest decreases, whereas Tasmania (-1.5 MW/mm) shows the smallest. However, every model shows a R^2 below 0.004, indicating that rainfall explains less than 0.5 % of total demand variability. Thus, precipitation's impact is minimal and largely overshadowed by other temporal factors such as temperature, time of day, and season.

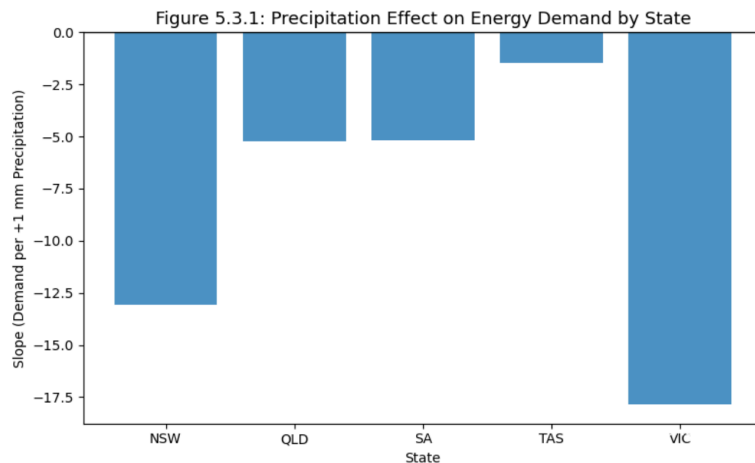


Table 5.3.2: Precipitation effect on Energy Demand per state

Figure 5.3.2 shows that rainfall is linked to slight decreases in electricity demand across all states. Victoria and New South Wales record the steepest declines, while Tasmania shows the weakest. Although consistent, these effects are minimal, with R^2 values under 0.004, indicating that

precipitation has only a minor, indirect influence—mainly due to cooler, cloudier conditions that lower air-conditioning use.

5.4 OVERALL MODEL & FORECAST MODEL

This section evaluates the performance of the overall Random Forest model developed to predict total electricity demand across Australia. The model incorporated temporal features (hour, month, year, weekend indicators), weather variables (temperature, humidity, precipitation), and state identifiers. The objective was to capture both the short-term daily fluctuations and long-term seasonal patterns in energy consumption, first by capturing a R^2 reading by comparing to past data, and then using this model to forecast the predicted Demand.

5.4.1 MODEL DEVELOPMENT

To forecast electricity demand, a **Random Forest Regressor** was trained using hourly energy and weather data from 2000–2018, and evaluated on the 2019 test period. The goal was to assess how well temporal and weather features could capture demand fluctuations across Australian states. Once a reliable correlation with our factors and demand have been confirmed, this model was then forecasted to predict future demand in the year to come.

Data Preparation

The model used the cleaned dataset containing total electricity demand (MW), air temperature (°C), and state identifiers. Datetime information was converted into structured time features to capture temporal patterns with: **Hour, Month, Year, isWeekend & State**.

Training and Testing Split

A time-based split was applied to the master data:

- **Training set:** all data up to **31 December 2018**.
- **Testing set:** all data from **1 January 2019-31 December 2019**.

This approach was specifically chosen to stimulate real-world forecasting, where models are trained on historical data to predict future demand (**Ma, L. (2019)**).

5.4.2 MODEL CREATION

A **Random Forest Regressor** was chosen for its robustness and ability to model non-linear relationships. The model was fitted on the training features and evaluated using R^2 , Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE) on both train and test sets. Model evaluation showed high accuracy, achieving an **R^2 of 0.99** on training data and 0.98 on the 2019 test set, with corresponding **RMSE = 349 MW** and **MAE = 223 MW**. (figure 5.4.1). These results confirm that the model generalises effectively to unseen data and captures most of the variance in electricity demand.

Feature Importance

	R^2	RMSE	MAE
train	0.99	288.05	182.35
test	0.98	349.06	223.50

Figure 5.4.1. Model Evaluation Features

Top 10 features:	
State_TAS	0.441553
State_SA	0.341006
State_QLD	0.088748
Hour	0.045736
State_VIC	0.044434
Temperature	0.013313
Year	0.011887
IsWeekend	0.009247
Month	0.004077

Figure 5.4.2 Most Important Features from model

The **feature importance analysis** (Figure 5.4.2) revealed that **state identifiers** accounted for the largest share of predictive power, particularly Tasmania (44%), South Australia (34%), and Queensland (9%). This indicates strong regional differentiation in baseline demand levels which correlates with all previous information acquired in this report.

Temporal variables such as **Hour** (4.6%), **Month**, **Year**, and **Weekend indicator** also contributed meaningfully, reflecting daily and seasonal usage cycles. **Temperature** ranked as the highest impacting weather factor (1.3%), while **humidity** and **precipitation** contributed minimally, consistent with earlier exploratory findings that rainfall and humidity exert weak direct effects on electricity demand.

Overall, these values demonstrate that the Random Forest effectively models the variation in demand, achieving high accuracy while maintaining interpretability through its structure.

5.4.3 VISUALISATION

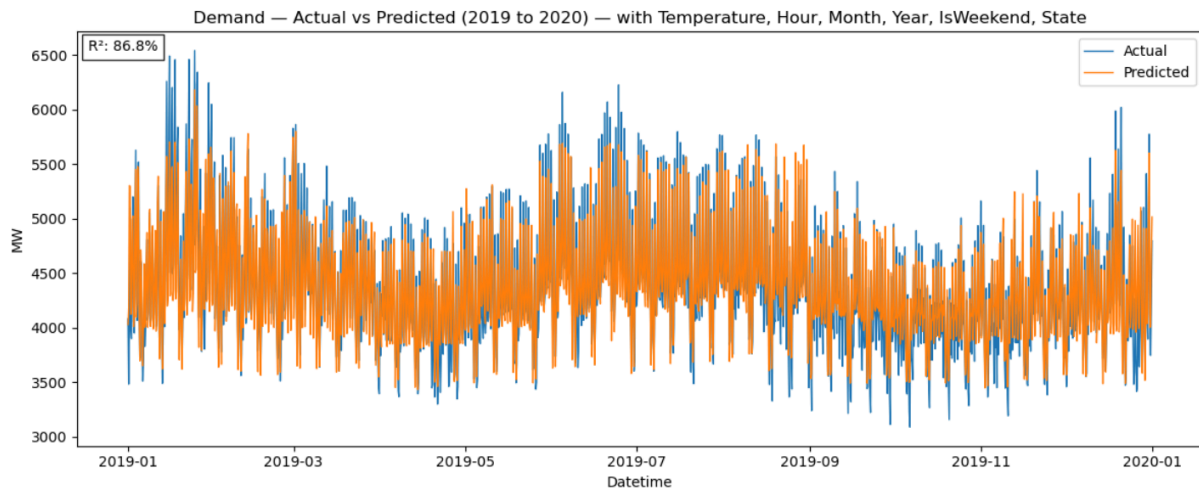


Figure 5.4.3: Actual vs Predicted Electricity Demand (2019 to 2020)

Figure 5.4.3 compares the model's predicted electricity demand (orange) against the observed actual demand (blue) for the 2019 test period. The Random Forest model, trained on data up to 2018, achieved a high R^2 of 86.8%, demonstrating strong alignment between predicted and real consumption patterns. The model successfully captures both daily and seasonal fluctuations—notably the summer and winter peaks—with a consistent overall trend throughout the year. However, it slightly underestimates extreme spikes in demand, such as those caused by heatwaves or abrupt cold snaps, being the main cause of the lower R^2 value than in the tested set.

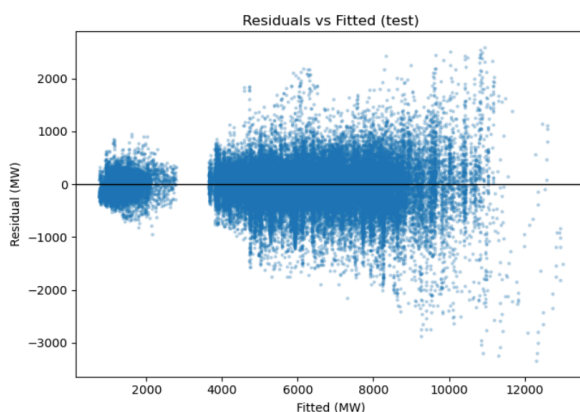


Figure 5.4.4: Residual graph of Overall model

Figure 5.4.4 plots residuals (the difference between actual and predicted demand) against the model's fitted values for the 2019 test set. The residuals are centered around zero, indicating that the model is unbiased overall and does not consistently overestimate or underestimate electricity demand. The spread of residuals increases slightly at higher fitted values, suggesting that the model performs less precisely during periods of very high demand, such as heatwaves or winter peaks, thus contributing to our previous analysis.

This validates the model's fit and demonstrates that the relationship between predictors and demand is appropriately modelled for the main range of values.

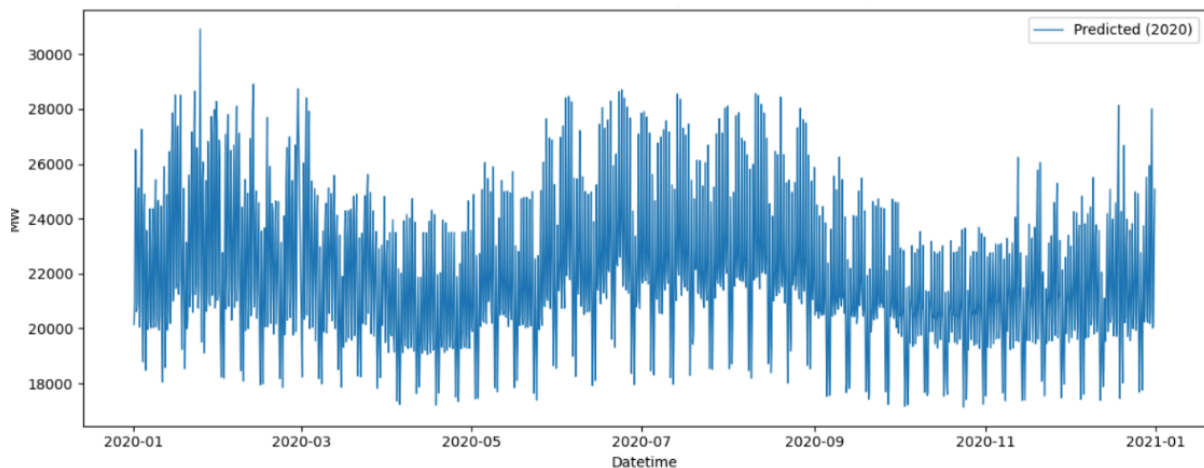


FIGURE 5.4.5 FORECAST OF ELECTRICITY DEMAND FOR 2020

Figure 5.4.5 presents the model's forecast of electricity demand for 2020, generated using weather conditions from the model trained on data up to 2018. The forecast model reproduces the expected seasonal U-shaped pattern, with higher demand during summer and winter months and lower demand through the milder spring and autumn months.

The model successfully captures daily and seasonal fluctuations, demonstrating strong generalisation beyond the training period. This result confirms the Random Forest model's ability to predict future demand trends, highlighting its usefulness for short to medium-term electricity forecasting and operational usage.

Summary

The Random forest model accurately predicted and forecasted electricity demand across Australia, achieving a performance of 86.6% on unseen 2019 data. Time-based variables and temperature were the main factors, while state differences explained regional demand variation. Humidity and precipitation had minimal influence, consistent with earlier findings. The model effectively captured daily and seasonal patterns but slightly underestimated extreme peaks.

Overall, this model demonstrates reliable generalisation and strong potential for short- to medium-term demand forecasting.

6 CONCLUSION

This project investigated how key climatic variables (such as air temperature, time, relative humidity and precipitation) affect the energy demand across East Coast Australian States between 2000 and 2019. The results from exploratory data analysis and modelling consistently show that temperature and time are the dominant factors shaping national energy consumption patterns, whereas humidity and precipitation have weaker, secondary effects.

Energy demand peaks during extreme temperature seasons such as winter and summer due to higher heating and cooling needs, and remains lowest during milder seasons of

spring and autumn when temperatures are more comfortable. Daily patterns showed strong human behavioural influence, with demand being lowest overnight (~4am) and highest in the early evening (~6pm). Electricity demand was also notably higher on weekdays, driven by workplace and industrial activity.

From the modelling phase, linear and quadratic regression models confirmed that air temperature is the strongest predictor of energy demand, displaying a U-shaped relationship with high demand at low and high temperatures extremes. When humidity was modelled together with temperature, its effect became easier to see. Humidity's effect on energy demand increases as temperature rises, suggesting that high humidity amplifies thermal discomfort and reduces air conditioning efficiency, making hot and humid days particularly energy intensive.

The overall Random Forest model achieved an R^2 value of 86.8%, indicating strong predictive power.

The findings provide several practical insights for energy providers and grid operators:

- Forecasting and planning: Utilise short term demand forecast models to anticipate peak load periods during heatwaves and cold snaps.
- Operational efficiency: Schedule maintenance and staffing during the weekend or mild weather periods (15-25°C) when demand is typically lower.
- Demand response: Implement demand response programs to manage the combined effects of extreme heat and high humidity.

Future research could extend this analysis by including regional and household level data, allowing for regional pattern analysis instead of limiting to overall state averages. Expanding coverage to include WA and the NT would provide a more complete national view. Incorporating other climatic factors such as wind speed and renewable energy data could also improve forecasting accuracy. More advanced models such as neural networks could capture more complex patterns and better predict extreme weather events.

Overall, the project demonstrates that while humidity and precipitation play minor roles individually, their combined effects with temperature and time significantly shape Australia's energy demand patterns. Understanding these dynamics is crucial for developing more reliable and efficient energy systems.

7 REFERENCES

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8 APPENDIX

Group Project Assessment for ADS1002
Studio: Tuesday 8am

Group: Energy 1							
		Member 1 Rohan	Member 2 xxxxx	Member 3 xxxxx	Member 4 xxxxx	Member 5 xxxxx	Member 6 xxxxx
Coordination	The sum of this row should be 100. The entry should reflect the group member's contribution to this project task.	24	24	24	24	4	0
Research	The sum of this row should be 100. The entry should reflect the group member's contribution to this project task.	25	25	25	25	0	0
Coding	The sum of this row should be 100. The entry should reflect the group member's contribution to this project task.	28	24	24	24	0	0
Report Writing	The sum of this row should be 100. The entry should reflect the group member's contribution to this project task.	20	20	20	30	10	0
Presentation Preparation	The sum of this row should be 100. The entry should reflect the group member's contribution to this project task.	23	27	18	27	5	0

Attendance at Final Presentation	This entry should be 1 (present) or 0 (absent)	1	1	1	1	1	0
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